Experiment 13: Preparation of standard sodium carbonate solution

Analysis of acids and bases often requires the determination of an accurate concentration of a solution. The wine industry for example relies on pH testing using titrations during the fermentation process. The grape juice 'must' is titrated with a standard solution of a base such as sodium hydroxide. Standard solutions are solutions whose concentrations are accurately known. Volumetric techniques rely on accurate measurement of quantities. Mass and volume are the usual quantities measured. This experiment is the first in a series that prepares some standard solutions. Once prepared the standard solutions can be used to determine the concentrations and pH of unknown samples such as a wine.

Notes

Anhydrous sodium carbonate of analytical reagent (A.R.) quality can be used as a primary standard as it is very pure and does not readily pick up moisture from the air. By dissolving a precisely known mass of Na_2CO_3 in a definite volume of solution it is possible to prepare a standard solution of Na_2CO_3, that is, one whose concentration is known accurately.

In this experiment you will prepare 500 mL of approximately 0.05 mol L^{-1} Na_2CO_3 solution whose concentration is accurately known.

Equipment

balance
volumetric flask (500 mL)
oven
desiccator
beaker (250 mL)
washbottle
storage bottle (approximately 500 mL)
distilled water
anhydrous sodium carbonate [Na_2CO_3] (4 g)

Procedure

1. Calculate the mass of anhydrous Na_2CO_3 required to make up 500 mL of 0.05 mol L^{-1} solution.
2. Place a little more than the required amount in an oven at 270 °C for 30 minutes to remove any water. After drying, place the anhydrous Na_2CO_3 in a desiccator to cool.
3. Accurately weigh out into a 250 mL beaker a mass of Na_2CO_3 approximately equal to that calculated. You should not waste time trying to weigh out exactly the mass calculated but the mass must be known precisely so that the exact concentration can be calculated.
4. Dissolve the solid in about 100 mL of distilled water.
5. Transfer this solution to a 500 mL volumetric flask. Rinse the beaker several times with about 20 mL portions of distilled water, adding each washing to the volumetric flask.
6. Make up the solution to precisely 500.0 mL with distilled water, adding the last few millilitres dropwise from a washbottle.

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Notes

7. Place the stopper in the volumetric flask and mix the solution thoroughly by repeatedly inverting the flask.

8. Transfer the solution to a clean storage bottle which should first be rinsed with a little of the Na_2CO_3 solution. Label the storage bottle with the type of solution and its date of preparation.

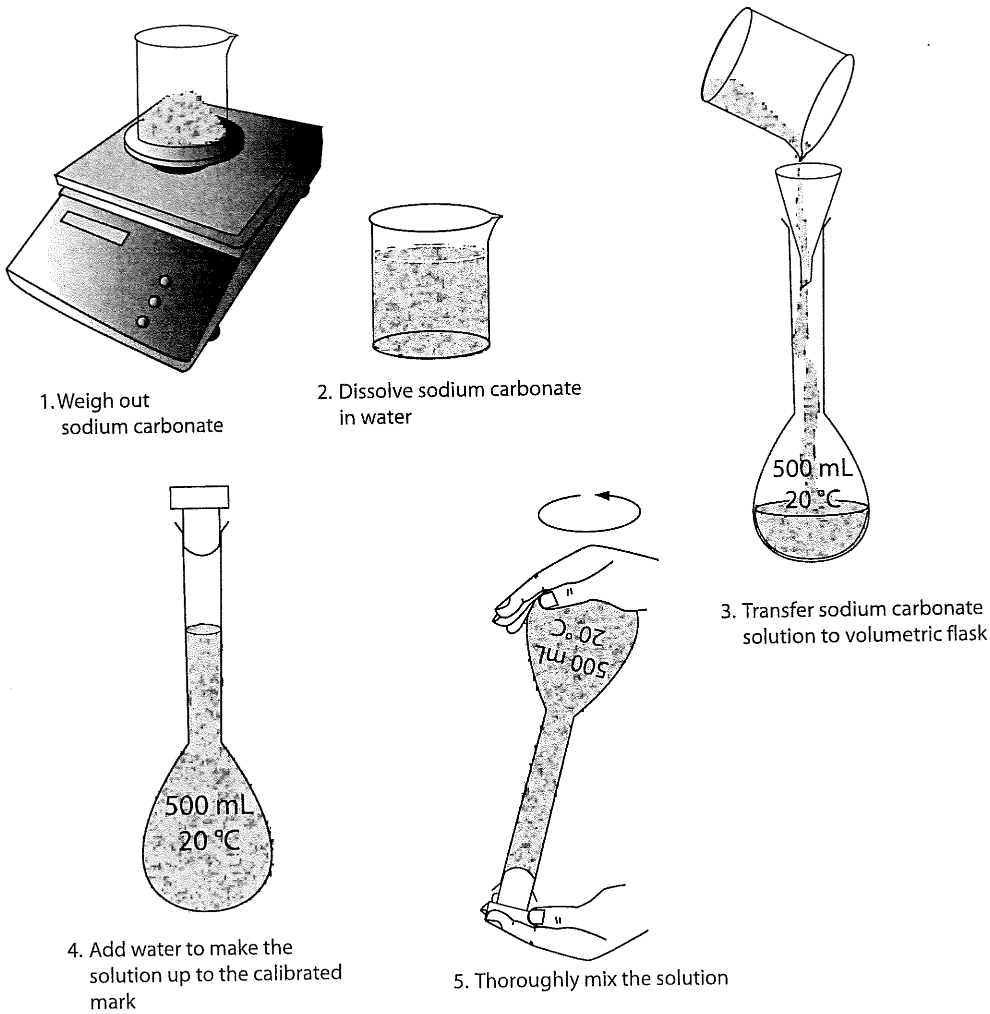


Figure 13.1: Preparing standard sodium carbonate solution

Processing of results and questions

1. Calculate the precise solution concentration in mol L^{-1}. Add this information to the label.

2. The concentration of the solution you have made should be accurately known. List some possible ways that could contribute to minor inaccuracies in determining the concentration.

Experiment 14: Preparation and standardisation of hydrochloric acid solution

Hydrochloric acid cannot be used as a primary standard. In this experiment you will prepare an approximately 0.1 mol L^{-1} solution and determine its exact concentration by titration against the standard Na_2CO_3 solution prepared in Experiment 13. The equation for the reaction is:

$$2H^+(aq) + CO_3^{2-}(aq) \leftrightharpoons H_2O(\ell) + CO_2(g)$$

Note that one mole of Na_2CO_3 reacts with two moles of $HC\ell$.

Since the solution at the equivalence point is somewhat acidic (pH ~ 3.5), an indicator that changes colour in this region must be used. Methyl orange (yellow to red when pH changes from 4.4 to 3.1) or bromophenol blue (blue to yellow for pH change from 4.6 to 3.0) are most suitable.

Notes

Equipment

concentrated hydrochloric acid [$HC\ell$] (6 mL)
graduated cylinder (10 mL)
volumetric flask (500 mL)
storage bottle (approximately 500 mL)
beakers (two 100 mL)
burette and stand
funnel
pipette (20 mL)
pipette filler
conical flask (250 mL)
standard sodium carbonate solution [Na_2CO_3] approximately 0.05 mol L^{-1} from Experiment 13 (150 mL)
methyl orange or bromophenol blue (a few drops)
distilled water

Procedure

Part A: Making the approximately 0.1 mol L^{-1} hydrochloric acid

SAFETY NOTE:
- **Concentrated hydrochloric acid is very corrosive and must be handled with extreme care.**
- **If any concentrated $HC\ell$ gets in contact with your skin, immediately wash it off with *copious* quantities of water.**

1. Calculate the volume of concentrated (approximately 12 mol L^{-1}) $HC\ell$ that would be required to prepare 500 mL of 0.1 mol L^{-1} solution.

2. Measure out this volume of concentrated $HC\ell$ in a graduated cylinder and transfer to a 500 mL volumetric flask that is about half filled with distilled water. Make the solution up to the mark.

3. Place the stopper in the volumetric flask and mix the solution thoroughly by repeatedly inverting the flask.

Experiment 14: Preparation and standardisation of hydrochloric acid solution

Notes

4. Transfer the approximately 0.1 mol L^{-1} $HC\ell$ to a clean storage bottle that has been rinsed with a little of the $HC\ell$ solution.

5. Label the storage bottle with the type of solution and the date of preparation.

Procedure

Part B: Standardisation of the hydrochloric acid solution

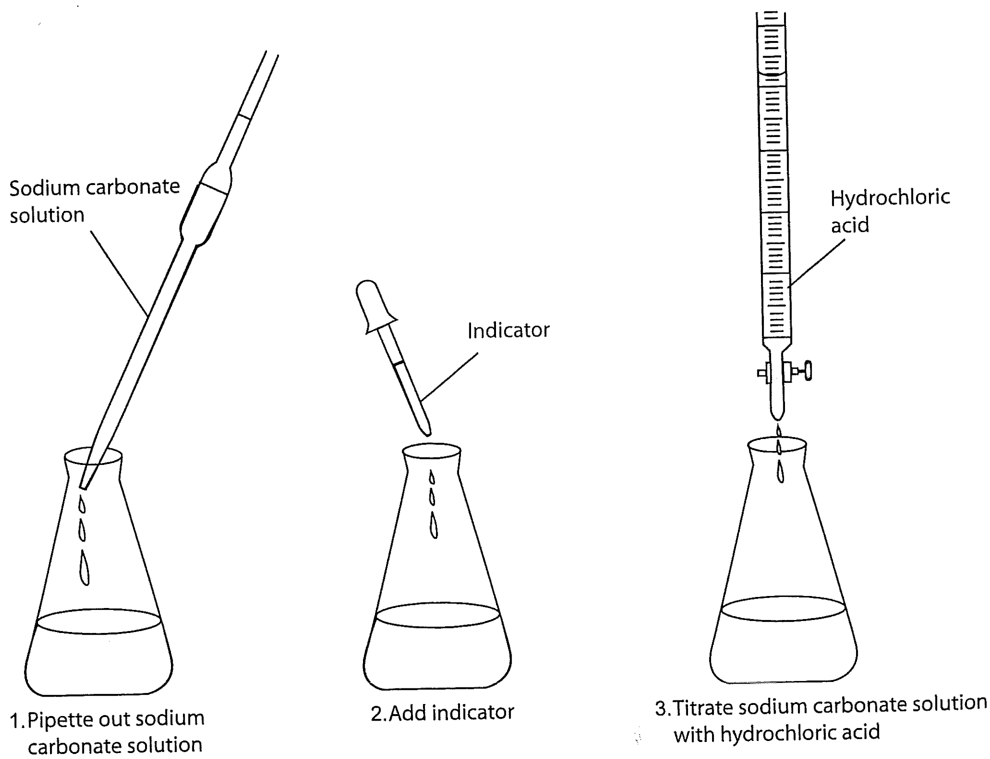
1. Place about 100 mL of the standard Na_2CO_3 solution into a clean beaker. If the beaker is wet, rinse with a little of the Na_2CO_3 solution first.

2. Rinse a clean 20 mL pipette with some of the Na_2CO_3 solution. Pipette 20 mL of the Na_2CO_3 solution into a 250 mL conical flask. Add 2-3 drops of your chosen indicator to the flask.

3. Place about 100 mL of $HC\ell$ into a clean beaker. Again, if necessary, rinse the beaker with a little of the $HC\ell$ solution first

4. Rinse a clean burette with some of the $HC\ell$ and then fill the burette with the solution.

5. Note and record the level of acid in the burette. Obtain a rough estimate of the titration volume by running acid quickly from the burette while constantly swirling the liquid in the conical flask. Stop delivery of the acid as soon as a permanent colour change is obtained. Note and record the acid level in the burette and determine the approximate volume of acid required.



6. Record your results in a table similar to the one shown below.

	Rough estimate	Accurate titrations		
		1	2	3
Final reading (mL)				
Initial reading (mL)				
Titration volume (mL)				

7. Prepare another conical flask containing 20 mL of the Na_2CO_3 solution and 2-3 drops of indicator. This time add the acid quickly from the burette with constant swirling of the flask, until the volume added is within 2-3 mL of the approximate volume required. Rinse the inside of the conical flask with a jet of water from a wash bottle to return any splashed solution to the bulk. Continue adding acid drop by drop, and with constant swirling, until the addition of one drop is sufficient to produce a permanent colour change. Note and record the level of the acid in the burette at the end point.

8. Repeat the accurate titration with further 20 mL portions of Na_2CO_3 solution until consistent titration volumes are obtained. These should be within 0.2 mL of each other.

Notes

Processing of results and questions

1. Using the equation for the reaction, calculate the number of moles of Na_2CO_3 used in each titration.

2. From the equation determine the number of moles of $HC\ell$ that react with each mole of Na_2CO_3 . Use this to determine the number of moles of $HC\ell$ used in the titration.

3. Calculate the average volume of $HC\ell$ used in the titrations. Use only those results that are reasonably consistent, that is within 0.2 mL of each other. From this determine the concentration of the $HC\ell$. Mark this information on the label of the storage bottle.

4. Distinguish between the `equivalence point' of a reaction and the 'end point' of a titration.

5. In the introduction it was stated that the solution at the equivalence point is 'somewhat acidic'. Explain why this is so.

6. Why is it that hydrochloric acid cannot be used as a primary standard?

7. Suppose that phenolphthalein, whose colour change is in the vicinity of pH 9, had been used instead of one of the indicators recommended.
   a) Would the volume of acid required for the titration be more or less than that obtained?
   b) Would the calculated concentration of the $HC\ell$ be higher or lower than the result obtained?

8. What are possible sources of error in this experiment?

Experiment 15: Preparation and standardisation of sodium hydroxide solution

Notes

A standard sodium hydroxide solution is often used to determine the acid content of wine.

Sodium hydroxide cannot be used as a primary standard as it readily absorbs water and carbon dioxide from the air. In this experiment you will make up a solution of NaOH that is approximately 0.1 mol L^{-1} and standardise this with standard approximately 0.1 mol L^{-1} $HC\ell$ prepared in Experiment 14.
The equation for the titration is:

$$\underset{\text{(from NaOH)}}{OH^-(aq)} \quad + \quad \underset{\text{(from HC}\ell\text{)}}{2H^+(aq)} \quad \leftrightarrows \quad H_2O(\ell)$$

At the equivalence point of the reaction, the pH changes from about 10 to 4 over the addition of only 0.1 mL of acid. Consequently any indicator that changes colour within this pH range is suitable.

Equipment

sodium hydroxide [NaOH] (3 g)
balance
beaker (250 mL)
volumetric flask (500 mL)
storage bottle (approximately 500 mL)
beakers (two 100 mL)
washbottle
burette and stand
conical flask (250 mL)
funnel
pipette (20 mL)
pipette filler
standard hydrochloric acid [$HC\ell$] approximately 0.1 mol^{-1} from Experiment 14 (150 mL)
methyl orange or phenolphthalein (a few drops)

Procedure

Part A: Making approximately 0.1 mol L^{-1} sodium hydroxide solution

1. Calculate the mass of NaOH needed to make up 500 ml, of 0.1 mol L^{-1} solution.

SAFETY NOTE:
- **Sodium hydroxide pellets are very corrosive and must not be allowed to come in contact with your skin.**
- **Use a spatula or plastic spoon to handle the NaOH pellets.**

2. Quickly weigh out this amount, to the nearest pellet, in a clean dry 250 mL beaker.

3. Dissolve the NaOH in about 100 mL of distilled water and transfer the solution to a 500 mL volumetric flask.

4. Stopper the flask and swirl the contents to mix them. If the solution is warm, wait until it cools, and then make it up to the graduation mark with distilled water. Place the stopper in the top and invert and swirl the flask several times to ensure that the water in the neck is mixed thoroughly with the solution in the bottom.

Notes

5. Transfer the approximately 0.1 mol L^{-1} NaOH to a clean storage bottle that has been rinsed with a little of the NaOH solution. Label the solution.

Procedure Part B: Standardisation of the sodium hydroxide solution

1. Using the appropriate technique pipette 20 mL of NaOH solution into a 250 mL conical flask. Add 2-3 drops of methyl orange or phenolphthalein.

2. With your standard $HC\ell$ from Experiment 14 in the burette do a rapid titration to obtain a rough estimate of the volume of $HC\ell$ needed to neutralise the 20 mL of NaOH solution.

3. Carry out repeat titrations, adding the acid more slowly near the end point, until consistent titration volumes are obtained, as in experiment 14. Record your results as before.

Processing of results and questions

1. From the equation for the reaction, the volume of NaOH used, and the volume and concentration of the standard $HC\ell$, calculate the concentration of the NaOH. Mark this on the label of the NaOH storage bottle.

2. What is meant by deliquescence? Why is a substance that is deliquescent unsuitable for use as a primary standard?

Experiment 16: Acetic acid content in vinegar

Notes

Commercial vinegar usually contains about 3-5% acetic acid (CH_3COOH) by mass. The purpose of this experiment is to determine the exact acetic acid content of a commercial brand of vinegar. This is achieved by titration with standard approximately 0.1 mol L^{-1} NaOH from Experiment 15.

Acetic acid reacts with hydroxide ion according to the equation:

$$CH_3COOH(aq) + OH^-(aq) \leftrightarrows CH_3COO^-(aq) + H_2O(\ell)$$

The pH at the equivalence point of this reaction is approximately 8.7, making phenolphthalein (colourless to pink when the pH changes from 8.3 to 10) a suitable indicator.

Equipment

balance
vinegar (40 mL)
volumetric flask (250mL)
washbottle
conical flask (250 mL)
pipettes (20 mL and 25 mL)
pipette filler
burette and stand
funnel
beakers (two 100 mL)
standard sodium hydroxide solution [NaOH] approximately 0.1 mol L^{-1} from Experiment 15 (150 mL)
phenolphthalein (a few drops)

Procedure

1. Determine the density of the vinegar by weighing out a known volume delivered from a pipette or by using a hydrometer.

2. Using a pipette place 25.0 mL of vinegar into a 250 mL volumetric flask. Make the volume up to precisely 250.0 mL with distilled water. Mix well by repeatedly inverting the volumetric flask.

3. Titrate the diluted vinegar solution from a burette against 20.0 mL portions of the standard NaOH solution each with 1-2 drops of phenolphthalein indicator. Record your results as before.

Processing of results and questions

1. Write the equation for the reaction of acetic acid with sodium hydroxide and calculate the number of moles of
   (a) NaOH in the 20.0 mL samples of standard NaOH solution
   (b) acetic acid from the average volume titrated.

2. Calculate the concentration of acetic acid in the diluted vinegar.

3. Determine the concentration of acetic acid in the undiluted vinegar.

Notes

4. Calculate the mass of acetic acid in 1000 mL of undiluted vinegar.

5. Use the density of the vinegar to calculate the mass of 1000 mL of vinegar.

6. Determine the mass of acetic acid per 100 g of vinegar, that is, the percentage by mass of acetic acid in the vinegar.

7. What volume of the 0.1 mol L^{-1} NaOH solution would be required in a titration with 20.0 mL of the undiluted vinegar, using phenolphthalein as indicator?

8. Suppose that methyl orange, which changes colour at about pH 3.7, had been used instead of phenolphthalein in your experiment. Would you expect the calculated percentage of acetic acid to be too high or too low? Explain.

9. What are the possible sources of error in this experiment?